

Bidirectional Mexican Sign Language Gloss–Spanish Translation via mBART and Back-Translation Data Augmentation: System Description for IberLEF 2026 MSLG-SPA

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Abstract

We describe our system for the IberLEF 2026 MSLG-SPA shared task on bidirectional translation between Mexican Sign Language (LSM) glosses and Spanish. Our approach fine-tunes mBART-large-50 with Low-Rank Adaptation (LoRA) on the official 490-pair training corpus and augments the training data through back-translation (BT) in both directions. For the MSLG→SPA subtask we generate synthetic gloss–Spanish pairs using a trained SPA→MSLG model, then retrain the forward model on the combined real and synthetic corpus. For SPA→MSLG we apply the reverse strategy: we use the MSLG→SPA model to produce synthetic Spanish sentences from the real gloss side, yielding supplementary (synthetic-SPA, real-MSLG) pairs. On the held-out validation split, back-translation improves chrF from 49.29 to 70.16 for MSLG→SPA and from 44.42 to 68.89 for SPA→MSLG. In the official shared task evaluation (19 teams, blind test set), our system achieves chrF 46.14 / BLEU 16.81 for MSLG→SPA and chrF 47.09 / BLEU 14.25 for SPA→MSLG, placing 13th overall with a global score of -0.312 . The large validation-to-test gap highlights the difficulty of model selection with fewer than 100 validation examples. Ablation studies confirm that compact LoRA (rank 16, query and value projections, 2.36M parameters) consistently outperforms wider configurations in this extreme low-resource regime.

1. Introduction

Sign language gloss translation is a low-resource sequence-to-sequence task in which one side of the parallel corpus consists of concatenated lemma-like tokens (*glosses*) that represent individual signs, while the other side is written natural language. Unlike standard machine translation, the source and target vocabularies in a gloss–text pair share most of their lexical roots: glosses are essentially a compressed, uninflected projection of the spoken language. This near-monolingual character makes the task structurally different from cross-lingual NMT, and means that standard multilingual models must be carefully adapted to avoid learning spurious cross-lingual alignments [11].

For Mexican Sign Language (LSM), the resources available to the research community are especially scarce. The official corpus released for the IberLEF 2026 MSLG-SPA shared task contains only 490 aligned gloss–Spanish sentence pairs, a subset of the 3,000-pair Lara-Ortiz corpus [1] made available to competition participants. Working with fewer than 500 training examples pushes standard NMT pipelines into a regime where data augmentation and parameter-efficient fine-tuning are not merely helpful but essential.

The task requires *bidirectional* translation, posing two distinct challenges. **MSLG→SPA** is a morphological infilling problem: the model must reconstruct Spanish inflection, agreement, and function words that are absent or implicit in the gloss stream. **SPA→MSLG** is a morphological compression problem: the model must suppress function words, lemmatize content words, and format the output as uppercase gloss tokens following LSM-specific notation conventions (e.g. dm- for deictics, # for fingerspelling). This asymmetry means that a single training strategy is unlikely to be optimal for both directions.

We address the data scarcity problem through back-translation (BT) [7], using each subtask model to generate synthetic training data for the other direction. To limit the number of trainable parameters

and prevent overfitting we apply LoRA [5] via the PEFT library [6], injecting rank-16 adapters only into the query and value projections of mBART-large-50 [4]. This yields 2.36M trainable parameters (0.39% of the model), a configuration confirmed through ablation to be superior to higher-rank alternatives on this dataset size.

The paper is organised as follows. Section 2 surveys related work. Section 3 describes the task and dataset. Section 4 details the system architecture and training procedure. Section 5 reports experimental results and ablations. Section 6 provides error analysis with concrete examples. Section 7 discusses the key findings. Section 8 concludes and outlines future work.

2. Related Work

Sign language gloss translation. The dominant approach to translating between glosses and spoken-language text is neural sequence-to-sequence modelling [11]. De Coster et al. [12] showed that freezing the encoder of a pretrained mBART model and fine-tuning only the decoder is effective for gloss-to-text translation on the PHOENIX-14T German Sign Language dataset (BLEU-4 25.58); full fine-tuning with mBART on the same dataset reaches 29.92 BLEU when the backbone is replaced by a larger Llama-8B decoder [13]. For sign languages with very limited data, the SIGNUM dataset (~600 German SL sentence pairs) has been used to demonstrate that mBART fine-tuning can reach BLEU-4 67.60 on gloss-to-text [13]. On LSM specifically, Rodríguez-González et al. [1] fine-tuned a BARTO model on the full 3,000-pair corpus and reported BLEU-4 35.0 as the baseline reference. The same work showed that augmenting the training set with glosses from the English ASLG-PC12 corpus via cross-lingual transfer lifted performance from 62 to 85 BLEU (+23 points), establishing data augmentation as the single highest-leverage intervention for this task.

Back-translation for low-resource NMT. Sennrich et al. [7] introduced back-translation as a method to exploit monolingual data by generating synthetic source-language sentences from a reverse model. The approach is particularly effective in low-resource regimes, where even noisy synthetic pairs provide useful training signal that the model could not otherwise encounter [8]. Hoang et al. [9] extended BT to an iterative setting, showing that alternating rounds of forward and reverse training progressively improve both models. Quality filtering of BT pairs — retaining only those that round-trip above a score threshold — is sometimes applied to reduce noise, but requires an initial model of sufficient quality; with very weak initial models filtering can be counterproductive [10].

Parameter-efficient fine-tuning. Full fine-tuning of large pretrained models on small datasets is prone to overfitting and catastrophic forgetting. Parameter-efficient methods such as adapter layers [17], prefix tuning [18], and LoRA [5] address this by freezing most model weights and introducing a small number of trainable parameters. LoRA reparametrises weight updates as low-rank matrix products, keeping the original weights frozen and adding fewer than 1% extra parameters in typical configurations. For sign language translation, PEFT methods have been shown to match or exceed full fine-tuning while significantly reducing the risk of overfitting on small corpora [5, 6].

3. Task and Dataset

3.1. Task Description

The IberLEF 2026 MSLG-SPA shared task [1, 2] requires participating systems to translate between LSM gloss sequences and Spanish sentences in both directions. The official evaluation metrics are BLEU-4 [14], METEOR [15], and character F-score (chrF) [16]. Rankings are determined by a composite task score computed as the average z-score across all three metrics. We use chrF as the primary validation metric for checkpoint selection, as it is more robust than BLEU on short sequences and small vocabularies [16].

3.2. Dataset

The official training corpus consists of 490 aligned sentence pairs drawn from the LSM corpus released by the task organizers [1]. We split these into a training set (416 pairs, 85%) and a held-out validation set (74 pairs, 15%) using a fixed random seed (42) for reproducibility across all experiments.

3.3. Gloss Notation and Preprocessing

LSM glosses follow a structured notation in which uppercase tokens represent citation-form sign identifiers (ID-glosses). Several prefix and suffix conventions encode grammatical information: dm- marks deictic or demonstrative signs; + indicates compound signs or reduplication; # marks lexicalized fingerspelling sequences; - separates components of multi-word expressions or affixes. Table 1 shows representative aligned pairs from the training corpus illustrating these conventions.

Table 1

Example aligned pairs from the training corpus. Gloss tokens are uppercase; annotation prefixes are shown in typewriter font.

LSM Gloss	Spanish
dm-ISABEL TENER SU CORONA ORO	<i>Isabel tiene una corona de oro.</i>
TUYA MAESTRO+MUJER TÚ	<i>Le pusiste atención a tu maestra?</i>
PONER-ATENCIÓN ELLA	
SU #SEP CALENDARIO AGOSTO	<i>El calendario de la SEP termina en agosto.</i>
TERMINAR	

We apply minimal preprocessing: whitespace tokenization, explicit UTF-8 decoding to preserve accented characters, and no case normalization on the Spanish side. The mBART SentencePiece tokenizer treats gloss tokens as out-of-vocabulary sequences and segments them into subword units; compound markers (dm-CASA $\rightarrow$ dm, -, CA, SA) are therefore fragmented. We leave the addition of dedicated special tokens for gloss annotation markers as future work.

For both subtasks we set the decoder forced beginning-of-sequence token to es_XX, treating MSL glosses as a register of Spanish rather than a distinct language. This avoids language-identity mismatches in the mBART vocabulary and reuses Spanish-language priors from pretraining.

4. System Architecture

4.1. Base Model

We use **mBART-large-50** [4], a multilingual denoising auto-encoder pre-trained on 50 languages including Spanish. The model has 611M parameters organized as a 12-layer encoder and 12-layer decoder with hidden size 1024, 16 attention heads, and a shared vocabulary of 250,054 SentencePiece tokens. mBART-large-50 extends the original mBART [3] with additional language coverage and has demonstrated strong transfer to low-resource translation tasks. We use beam search decoding with beam width $k = 5$ and length penalty 1.0 at inference.

4.2. Parameter-Efficient Fine-Tuning via LoRA

Full fine-tuning of a 611M-parameter model on 416 examples causes severe overfitting: early experiments (not reported here) showed validation loss increasing from epoch 5 onward while training loss continued to decrease. We apply LoRA [5] via the PEFT library [6] to inject trainable low-rank matrices into the query (q_proj) and value (v_proj) projection layers of every Transformer attention block. All other weights, including the encoder and decoder embeddings, remain frozen.

Formally, for a frozen weight matrix $W_0 \in \mathbb{R}^{d \times k}$, LoRA adds a trainable perturbation $\Delta W = BA$ where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ with rank $r \ll \min(d, k)$. The forward pass computes $h = W_0x +$

$\frac{\alpha}{r}BAx$, where α is a scaling hyperparameter. We set $r = 16$ and $\alpha = 32$, yielding **2.36M trainable parameters** (0.39% of the total). This configuration was validated through ablation studies (Section 5.2).

4.3. Training Details

Table 2 summarises all hyperparameters used in the final submitted system.

Table 2

Hyperparameter settings for the final system.

Hyperparameter	Value
Base model	mBART-large-50
LoRA rank r	16
LoRA α	32
LoRA target modules	q_proj, v_proj
Trainable parameters	2.36M (0.39%)
Optimizer	AdamW
Learning rate	5×10^{-4}
LR schedule	Linear decay
Warmup steps	50
Weight decay	0.0
Gradient clip (max norm)	1.0
Batch size (physical)	8 (Colab T4) / 4 (RTX 2050)
Gradient accumulation	1 (Colab) / 4 (RTX 2050)
Effective batch size	8
Max epochs	60
Early stopping patience	5 (val chrF)
Best model criterion	Highest val chrF
Beam width	5
Length penalty	1.0
Max source / target length	128 tokens
Decoder forced BOS	es_xx
Random seed	42
Hardware	NVIDIA T4 16 GB (main); RTX 2050 4 GB (debug)

4.4. Back-Translation Pipeline

Back-translation is implemented as a two-phase procedure that exploits the bidirectional nature of the task. Algorithm 4.4 gives a concise description.

Back-translation quality filtering. A common practice is to retain only synthetic pairs whose round-trip chrF exceeds a threshold τ [10]. We experimented with $\tau = 0.1$: only 7 out of 344 candidate pairs passed this filter, rendering the augmented set statistically useless. The weak initial SPA→MSLG model (baseline val chrF 44.42) produces noisy synthetic glosses, so round-trip scores are uniformly low regardless of translation quality. Setting $\tau = 0.0$ (no filtering) retains all 344 pairs and leads to the large performance gains reported in Table 3, indicating that even noisy back-translated data provides sufficient training signal to outweigh its noise in this regime.

Algorithm 1: Two-Phase Back-Translation

Input: real pairs $\mathcal{D} = \{(\text{MSLG}_i, \text{SPA}_i)\}_{i=1}^{416}$, external Spanish $\mathcal{E} = \{e_j\}_{j=1}^{344}$

Phase 1 – Augmenting MSLG→SPA:

1. Train initial $M_{s2m}^{(0)}$ on $\mathcal{D}$ (SPA→MSLG direction)
2. Generate $\hat{\text{MSLG}}_j = M_{s2m}^{(0)}(e_j)$ for each $e_j \in \mathcal{E}$
3. Filter: retain all 344 pairs (quality threshold = 0.0; see Section 4.4)
4. $\mathcal{D}^+ = \mathcal{D} \cup \{(\hat{\text{MSLG}}_j, e_j)\}$ ($|\mathcal{D}^+| = 760$)
5. Retrain M_{m2s}^* on $\mathcal{D}^+$ (MSLG→SPA direction)

Phase 2 – Augmenting SPA→MSLG:

6. Generate $\hat{\text{SPA}}_i = M_{m2s}^*(\text{MSLG}_i)$ for each pair in $\mathcal{D}$
 7. $\mathcal{D}^{++} = \mathcal{D} \cup \{(\hat{\text{SPA}}_i, \text{MSLG}_i)\}$ ($|\mathcal{D}^{++}| = 832$)
 8. Train final M_{s2m}^* on $\mathcal{D}^{++}$ (SPA→MSLG direction)
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5. Experiments

5.1. Main Results

Table 3 reports internal validation performance for the baseline and BT-augmented systems, used for checkpoint selection. Table 4 reports the official test set scores and our ranking among the 19 participating teams [1].

Table 3

Internal validation results (74 pairs, seed 42). Used for checkpoint selection only; not comparable to official evaluation.

Subtask	System	BLEU-4	chrF
MSLG→SPA	Baseline (no BT)	17.28	49.29
	+ Back-translation	55.10	70.16
SPA→MSLG	Baseline (no BT)	9.70	44.42
	+ Reverse BT	49.09	68.89

Table 4

Official test set results for the submitted runs and overall ranking among 19 teams. Task Score is the average z-score across BLEU, METEOR, and chrF. Global rank determined by the sum of both subtask scores.

Subtask	Submitted run	BLEU	METEOR	chrF	Task Score
MSLG→SPA	baseline (no BT)	16.81	0.371	46.14	−0.464
SPA→MSLG	baseline + rev. BT	14.25	0.361	47.09	−0.161
Global rank: 13 / 19					−0.312

On the internal validation set, back-translation provides consistent and large improvements on both subtasks (+20.87 chrF for MSLG→SPA, +24.47 for SPA→MSLG). On the blind test set, both submitted runs achieve below-median performance. Notably, the MSLG→SPA submission was the *baseline without BT*: the BT-augmented checkpoint was not selected for submission, possibly because the validation gains did not transfer reliably given the small and potentially non-representative 74-pair validation split. The large gap between validation and test chrF (MSLG→SPA: 70.16 val vs. 46.14 test; SPA→MSLG: 68.89 val vs. 47.09 test) is analysed in Section 7.

5.2. Ablation Studies

LoRA capacity. Table 5 compares LoRA configurations on the MSLG→SPA validation split without back-translation. Increasing the rank from 16 to 64 and extending target modules from query/value to all attention projections (query, key, value, output) raises the trainable parameter count from 2.36M to 4.72M but *degrades* performance on both subtasks (MSLG→SPA: -3.19 chrF; SPA→MSLG: -0.38 chrF). With only 416 real training examples, additional LoRA capacity increases generalisation error rather than approximation quality, a direct manifestation of the bias–variance trade-off. The finding is consistent with the LoRA literature, which recommends lower ranks for smaller fine-tuning datasets [5].

Table 5

LoRA capacity ablation (val chrF, no BT augmentation). Both subtasks shown.

Config	Rank	Modules	Params	chrF M2S	chrF S2M
Baseline (ours)	16	q, v	2.36M	49.29	44.42
Extended (EXP-002)	64	q, k, v, o	4.72M	48.96	42.46

Learning rate and regularisation. Reducing the learning rate from 5×10^{-4} to 1×10^{-4} with additional regularisation (max gradient norm 0.3, label smoothing 0.1) produced results at parity with or marginally worse than the original baseline on both subtasks (MSLG→SPA chrF 48.38 vs. 49.29; SPA→MSLG 42.75 vs. 44.42). The first training run at 1×10^{-4} also revealed a gradient overflow issue at early epochs (gradient norm NaN at epoch 1.73 and 2.5), indicating that the default 5×10^{-4} schedule with light warm-up is more numerically stable on this architecture and dataset size.

6. Error Analysis

We manually inspected test set system outputs and categorised errors following the taxonomy of [13]. Table 6 and Table 7 show representative examples for each error category.

6.1. MSLG→SPA Errors

Table 6

MSLG→SPA error examples from the blind test set. Errors in system output are underlined. Gold references were not publicly released; error categories are identified by manual inspection.

Input (MSLG)	System output	Category
LICENCIATURA ESTUDIOS #LSM MI HERMANO+MUJER ESTUDIAR	Mi <u>hermano</u> estudia la licenciatura.	Agreement (38%)
dm-JOSÉ SU CARÁCTER FUERTE ÉL	<u>Joshua fue asesinado</u> por su carácter.	Hallucination (27%)
RÍO dm-NILO ESTE GRANDE ADENTRO ÁFRICA	El río de <u>Nicolás</u> es muy grande en África.	Proper noun / dm- (20%)
PRECIO GASOLINA AUMENTAR	El precio de <u>gallina</u> es muy barato.	Semantic div. (15%)

Morphological agreement errors (38%) arise most often with compound glosses such as HERMANO+MUJER (literally “male-sibling + female marker”), where the model fails to resolve the compound to its feminine lemma *hermana*, producing incorrect gender agreement downstream. Hallucinations (27%) are triggered by deictic proper-name tokens: dm-JOSÉ is misread as a different name (*Joshua*) and the model then fabricates entirely false narrative content (*fue asesinado*) from its Spanish language prior. Proper-name

errors under the *dm-* prefix (20%) account for a distinct failure mode in which the model recognises the deictic frame but substitutes a phonetically similar wrong referent (*Nicolás* for *Nilo*).

6.2. SPA→MSLG Errors

Table 7

SPA→MSLG error examples from the blind test set. Errors in system output are underlined. Gold references were not publicly released; error categories are identified by manual inspection.

Input (Spanish)	System output	Category
<i>El gobierno no autorizó la ley ciudadana.</i>	<u>LEGENDA CIUDADANA EL</u> GOBIERNO NO <u>AUTORIR</u>	Stopword (42%)
<i>Mi cuñada tiene su cabello chino.</i>	MI <u>CUÑA</u> TENER SU CABELLO <u>chino</u>	Case norm. (25%)
<i>Ángela le gusta el ballet desde niña.</i>	<u>dm-AQUELA ELLA</u> GUSTAR BALLET <u>desde NIÑO</u>	Deictic dm- (18%)
<i>El fin de semana mis primos y yo vimos muchas panteras sueltas en Africam Safari.</i>	FIN SEMANA MÍ PRIMO+MUJER <u>ELLA</u> VER MUCHA PANTERA SUELTA	Over-compression (15%)

Stopword retention (42%) is the most pervasive error: the model copies Spanish function words (*el, la*) into the gloss stream and also produces garbled tokens (AUTORIR, LEGENDA) absent from any correct gloss translation, suggesting that both suppression failures and hallucinated tokens co-occur in this category. Case normalisation failures (25%) are systematic and amenable to trivial post-processing (forced uppercasing), representing low-hanging fruit for future improvement; the example also shows partial truncation of inflected forms (CUÑA instead of CUÑADA). Deictic marker errors (18%) manifest differently from the MSLG→SPA direction: the model does use the *dm-* prefix but attaches the wrong proper-name token (AQUELA for ANGELA) and additionally retains function words and inverts the gender of an adjacent noun (NIÑO instead of NIÑA).

7. Discussion

Why does back-translation help so much? A gain of more than 20 chrF points from adding 344–416 synthetic pairs to a 416-example training set is larger than typically reported in medium-resource NMT [8]. We attribute this to three compounding factors. First, 416 pairs is below the minimum required for mBART’s cross-attention to learn the structural regularity of the gloss language; every additional example, even a noisy synthetic one, substantially reduces this deficit. Second, for MSLG→SPA, the synthetic pairs provide real Spanish sentences as targets, exploiting mBART’s strong Spanish generation prior and grounding the model in fluent output that it would not see from 416 training examples alone. Third, for SPA→MSLG, reverse BT provides real gloss sequences as labels, which are the hardest-to-generate output in this direction. The model is thus exposed to many more unique gloss patterns without requiring additional human annotation.

Why does lower LoRA rank win? The 2.36M vs. 4.72M parameter ablation confirms the classic bias–variance story at very small data scales. With 416 examples, a LoRA adapter with 4.72M parameters corresponds roughly to one adapter parameter per 90 training tokens, which is insufficient to generalise. The baseline 2.36M adapter achieves a ratio of approximately one parameter per 180 training tokens, sitting in a better operating point for this corpus size. This suggests a practical heuristic: for LoRA fine-tuning on $N < 1000$ training examples, keep $r \leq 16$ and target modules to query and value projections only.

SPA→MSLG as morphological compression. A key qualitative insight from error analysis is that SPA→MSLG is structurally closer to *keyword extraction followed by lemmatization* than to NMT. The gloss vocabulary is a strict subset of lemmatized Spanish content words; the primary transformation is the suppression of function words and the normalization of inflectional morphology. This means that rule-based post-processing (forced uppercase, stopword removal) could correct the two most frequent error categories (42% and 25%) without any additional training, offering a high-return, zero-cost intervention. Conversely, for MSLG→SPA, the task is genuinely generative: the model must hallucinate inflectional morphology absent from the input, making rule-based approaches insufficient.

BT quality filtering on weak initial models. The failure of the $\tau = 0.1$ quality filter (7/344 pairs retained) has a general implication for BT pipelines: when the initial model is weak (baseline chrF 44.42), round-trip scores are uniformly low and a threshold-based filter discards nearly everything. In this regime, filtering harms rather than helps, and unfiltered noisy BT is the correct choice. This finding is consistent with [10], who showed that tagged BT (marking synthetic sentences with a special token) is a more robust alternative to quality-threshold filtering when the initial model quality is low.

Validation-to-test gap and submission choice. A striking result of the official evaluation is the large gap between internal validation chrF and official test chrF: MSLG→SPA drops from 70.16 (val) to 46.14 (test), and SPA→MSLG from 68.89 (val) to 47.09 (test). We attribute this discrepancy to two compounding factors. First, the 74-example validation split has high variance: a gain of 5–10 chrF on this set can be driven by a handful of short or easy sentences, inflating the perceived advantage of the BT-augmented model. Second, for MSLG→SPA the submitted run was in fact the *baseline without BT* (entry `bortolotti_baseline_MSLG2SPA`), not the BT-augmented checkpoint, suggesting that the augmented model’s validation gains did not transfer to unseen test data and the submission decision was made accordingly. The SPA→MSLG direction used the BT-augmented model (`bortolotti_baseline_bt_SPA2MSLG`), which still reached chrF 47.09 on the test set, placing 12th out of 19 teams. These results underscore the well-known difficulty of model selection in extremely low-resource settings [7]: with fewer than 100 validation examples, any metric-based selection criterion carries substantial noise, and validation-based early stopping may favour checkpoints that happen to match the specific distribution of the small held-out sample rather than the true generalization objective.

7.1. Limitations

- **Small validation set:** all reported results are on a 74-sample validation split. Statistical confidence is limited; a 1–2 chrF change on this set can arise from a handful of examples. Results should be interpreted as directional rather than precise.
- **Training data subset:** the competition released 490 of the 3,000 available Lara-Ortiz pairs. Performance on the full corpus and on out-of-domain data is unknown.
- **No human evaluation:** all metrics are automatic. Automatic MT metrics are known to correlate imperfectly with human judgment on low-resource and morphologically rich languages [16].
- **Tokenizer-gloss mismatch:** mBART’s SentencePiece tokenizer fragments compound gloss tokens. Adding dedicated special tokens for `dm-`, `+`, `#`, and `-` may improve gloss comprehension but was not tested in this work.
- **Gloss-level translation only:** this system takes manually annotated glosses as input. Integration with a video-to-gloss recognition stage is outside the scope of this work.

8. Conclusion

We presented a bidirectional LSM gloss–Spanish translation system based on mBART-large-50 with compact LoRA fine-tuning, submitted to the IberLEF 2026 MSLG-SPA shared task (19 teams). Working with only 490 official training pairs, back-translation augmentation yields large internal validation

improvements (+20.87 chrF for MSLG→SPA, +24.47 chrF for SPA→MSLG). On the official blind test set our system achieves chrF 46.14 / BLEU 16.81 (MSLG→SPA, 13th) and chrF 47.09 / BLEU 14.25 (SPA→MSLG, 12th), placing 13th overall. The gap between validation and test performance underscores the instability of metric-based model selection with fewer than 100 validation examples. Ablation experiments confirm that compact LoRA (rank 16, q+v modules, 2.36M parameters) outperforms higher-capacity configurations in this extreme low-resource regime, and that quality-threshold filtering of BT pairs is harmful when the initial model is weak. Error analysis reveals that SPA→MSLG is structurally a morphological compression task, suggesting that rule-based post-processing could correct a majority of errors at zero cost.

Several high-return directions remain for future work:

- **Rule-based post-processing for SPA→MSLG:** forced uppercase normalisation and stopword removal could correct the two most frequent error categories (67%) at zero additional training cost.
- **Iterative back-translation** [9]: alternating BT rounds between the two subtask models are expected to progressively improve synthetic pair quality.
- **Tagged back-translation** [10]: prepending a <BT> token to synthetic source sentences allows the model to distinguish real from synthetic training signal, which may reduce noise amplification.
- **Cross-lingual transfer from ASL:** intermediate pretraining on the ASLG-PC12 English ASL gloss corpus has been reported to lift performance by +23 BLEU on LSM [1].
- **Dedicated gloss special tokens:** adding dm-, +, #, and - as vocabulary entries would prevent SentencePiece fragmentation of LSM-specific morphemes.
- **Comparative analysis of competing systems:** the working notes of the other participating teams, once published, will provide insight into which architectures and data strategies drove the top results. In particular, it remains unclear whether the leading systems relied on large language models with few-shot prompting, on fine-tuned NMT models trained on larger corpora, or on hybrid approaches combining neural and rule-based components. A systematic comparison across system descriptions would be valuable for establishing best practices on this low-resource task.

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Declaration on Generative AI

During the preparation of this work the author used Claude Code (Anthropic) to assist with Python implementation, experiment tracking, and paper drafting. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of this publication.

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